PHANTOM PHOENIX: Digital Media Authenticity & AI Forensic Intelligence Platform

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***Abstract*—We present PHANTOM PHOENIX, an end-to-end platform for AI-generated image detection and forensic screening. The system integrates a ViT-Small-based detector trained on the Community Forensics dataset (2.7M images from 4,803 generators) with complementary forensic analyzers: EXIF/metadata extraction, Error Level Analysis (ELA), noise-residual statistics, and 2D FFT frequency-spectrum analysis. Evidence from the detector and forensic analyzers is normalized into a unified schema distinguishing `model_inference`, `computed_observation`, `heuristic`, `verified_evidence`, and `unknown_or_unavailable` categories, enabling transparent, auditable verdicts. A deterministic verdict engine applies configurable thresholds (`likely_ai` $\geq$ 0.80, `likely_real` $\leq$ 0.20) with explicit `not_claimed` disclaimers (no generator identity, legal authenticity, localized forgery, or calibrated confidence claims). On the locked 922-sample test split (452 real, 470 AI-generated) from the PHANTOM PHOENIX 6,000-image manifest (stratified 70/15/15, zero hash overlap), the frozen validation-selected threshold (0.05) yields accuracy 0.7538, precision 0.9765, recall 0.5298, F1 0.6869, ROC-AUC 0.8959, with a false positive rate of 0.0133 (6 false positives out of 452 real images). The platform provides authenticated scan persistence, audit logging, and publication-ready PDF forensic dossiers. We report verified system-level functionality; scientific detector evaluation remains bounded by dataset provenance and threshold-selection provenance as documented.**

***Index Terms*—AI-generated image detection, deepfake detection, digital forensics, vision transformer, forensic analysis, evidence aggregation, explainable AI, Community Forensics, OpenFake**

1. Introduction

The proliferation of text-to-image diffusion models has made photorealistic synthetic image generation accessible at unprecedented scale. This capability challenges the evidentiary value of digital imagery in journalism, law enforcement, and public discourse. Existing detection benchmarks often rely on outdated generators or narrowly scoped datasets, limiting real-world utility [1], [2].

PHANTOM PHOENIX addresses this gap as a forensic *screening* platform rather than a definitive classifier. Its design principles are:

- **Transparency:** Every verdict traces to explicit evidence items with method tags and limitations.
- **Scope honesty:** Verdicts carry mandatory `not_claimed` disclaimers; the system never asserts generator identity, legal authenticity, or localized forgery.
- **Multi-modal evidence:** AI detector output is presented alongside classical forensic signals (ELA, noise, frequency, metadata).
- **Auditability:** Every scan is persisted with full provenance; authentication and audit trails are tamper-evident.
- **Reproducibility:** Model revision, preprocessing, dataset splits, and evaluation protocols are pinned and documented.

The deployed detector is `buildborderless/CommunityForensics-DeepfakeDet-ViT` (ViT-Small, revision `ac6ee457bea904a37306575410745179 3b56`) the official Hugging Face `transformers` port of the Community Forensics CVPR 2025 checkpoint [1]. An earlier EfficientNet-B4 prototype was evaluated but not deployed; the production system uses ViT-Small exclusively.

2. Related Work

AI-Generated Image Detection.. Early detectors targeted GAN artifacts [3], [4]. Modern diffusion models produce fewer low-level artifacts, shifting focus to representation-level generalization. The Community Forensics work [1] demonstrated that training on thousands of generators (4,803 models, 2.7M images) yields substantially better out-of-distribution generalization than smaller datasets. Our deployed ViT-Small detector is the direct product of that training regime.

Benchmarks and Datasets.. GenImage [5] and DiffusionDB [6] provided early diffusion-era benchmarks. OpenFake [2] introduced a politically grounded dataset with frontier proprietary generators and a crowdsourced adversarial platform, achieving near-perfect in-distribution F1 (0.997)

and strong OOD generalization. Our 6,000-image evaluation manifest draws from the OpenFake `core` configuration (train/val/test splits) with held-out OOD generators in test.

Classical Forensic Analyzers.. Error Level Analysis (ELA) [7] detects JPEG recompression inconsistencies. Noise-residual statistics [8] capture sensor-pattern deviations. Frequency-domain analysis [9] reveals generator-specific spectral signatures. These methods are heuristic or computed observations, not calibrated detectors; we present them as complementary signals with explicit method tags.

Evidence Fusion and Explainability.. Prior works fuse multiple detectors [10], [11] or use attention visualization [12]. Our architecture keeps AI-generation assessment and manipulation indicators strictly separated, normalizing all evidence into a schema with five evidence categories and a deterministic verdict engine.

3. Problem Definition and Scope

Input: A single image file (JPEG, PNG, WEBP; $\leq$15 MB, $\leq$8192 px on shortest edge). **Output:** A structured forensic assessment containing:

1) AI-generation probability from the primary detector (ViT-Small).
2) Forensic indicator results: metadata, ELA, noise, frequency.
3) Normalized evidence items with method categories and limitations.
4) A deterministic verdict: `LIKELY_AI`, `LIKELY_REAL`, `INCONCLUSIVE`, or `ANALYSIS_FAILED`.
5) Risk level: `LOW`, `MODERATE`, `HIGH`, or `UNKNOWN`.
6) Rationale traceable to evidence items; explicit `not_claimed` disclaimers.
7) Explainability artifact (CLS-to-patch attention rollout metrics) when available.
8) Persisted scan record with SHA-256, timestamp, owner, and artifact references.
9) Downloadable PDF forensic dossier.

Out of scope: Generator attribution, localized manipulation detection, watermark extraction, legal evidentiary certification, video/deepfake audio, real-time streaming analysis.

4. System Architecture and Evidence Model

4.1. High-Level Architecture

The platform is a three-tier web application:

- **Frontend:** Single-page React 18 application (TypeScript, Vite, hash-based routing) served via nginx. No external UI frameworks beyond `lucide-react` icons. Accessibility: focus-visible outlines, semantic landmarks, reduced-motion respect.
- **Backend:** FastAPI (Python 3.11) with async SQLAlchemy 2.0 / PostgreSQL. Stateless JWT authentication (HS256, Argon2id password hashing) delivered as `HttpOnly, SameSite=Lax` cookies. CSRF origin validation, per-socket-peer login rate limiting (5/min), in-memory sliding-window rate limits for scans (10/min) and default endpoints (120/min).
- **Database:** PostgreSQL 16 with asyncpg. Tables: `users`, `scans`, `audit_events`. Alembic migrations. Row-level ownership enforced; cross-user access returns 404 (no enumeration).

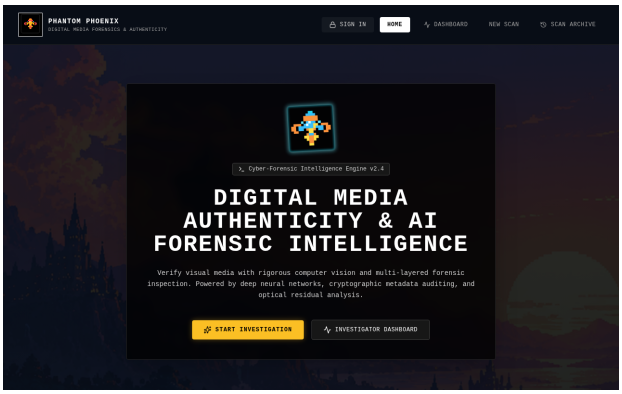


Figure 1: System entry point: landing page presenting the forensic workstation metaphor. The phoenix mascot denotes the investigative posture; amber primary CTA routes to authenticated upload.

Deployment: Docker Compose (dev) and `compose.staging.yaml` (TLS termination at `127.0.0.1:8443` via self-signed certificate, nginx proxy).

4.2. Evidence Normalization Schema

All detector and forensic outputs are mapped to `NormalizedEvidenceItem` with fields:

`id`	unique identifier (model ID or analyzer ID)
`source`	`ai_detector`, `forensic_metadata`,
`method`	`model_inference`, `computed`, `heuri`
`status`	`ok`, `error`, `not_applicable`, `preser`
`evidence_category`	`model_inference`, `computed_obser`
`result`	structured payload (probability, statistics, a
`interpretation`	human-readable explanation
`limitations`	string list from the producing component
`artifact`	relative path to generated forensic image (H

This schema enables the frontend to render structured cards instead of raw JSON (raw JSON remains accessible in a collapsible technical-details section).

4.3. Verdict Engine

The deterministic function `determine_verdict(model_outputs, likely_ai_threshold, likely_real_threshold)` implements:

TABLE 1: Forensic analyzers, methods, and evidence categories.

Analyzer	Method	Evidence Category	Status Values
AI Detector (ViT-Small)	model_inference	model_inference	ok, error
Metadata (EXIF/XMP)	computed	computed_observation	ok, present, absent
ELA (JPEG recompress q90)	computed	computed_observation	ok
Noise (median-filter residual)	heuristic	heuristic	ok
Frequency (2D FFT magnitude)	computed	computed_observation	ok

1) If primary model output missing, status != ok, or fake_probability is null → ANALYSIS_FAILED, Risk UNKNOWN.
2) Score $s \leftarrow$ primary fake_probability.
3) If $s \geq$ likely_ai_threshold (production 0.80) → LIKELY_AI, Risk HIGH.
4) If $s \leq$ likely_real_threshold (production 0.20) → LIKELY_REAL, Risk LOW (elevated to MODERATE if heuristic ELA/noise anomalies exceed internal thresholds).
5) Else → INCONCLUSIVE, Risk MODERATE.

Every verdict includes a rationale list (traceable to evidence items) and a fixed not_claimed list:

```
["generator identity",
 "legal authenticity",
 "localized forgery without verified provenance",
 "calibrated real-world criminal fraud probability"]
```

4.4. Forensic Analyzers

Metadata.. Extracts format, dimensions, color mode, EXIF tags (via PIL.ExifTags), XMP presence, software, datetime. Reports has_exif boolean and tag count.

ELA.. Recompresses image in-memory as JPEG quality 90; computes absolute difference; enhances by scaling to full dynamic range. Reports mean_difference, max_difference, std_difference on raw difference; saves enhanced ELA visualization artifact.

Noise.. Grayscale conversion; median filter (3×3) subtraction as high-pass noise residual estimate. Reports noise_mean, noise_std, noise_max; saves normalized noise map artifact.

Frequency.. Grayscale; 2D FFT with fftshift; log-magnitude spectrum. Reports mean_magnitude, std_magnitude, max_magnitude; saves normalized spectrum artifact.

Artifacts are saved under uploads/artifacts/{scan_id}/ and served via /api/artifacts/... with authentication.

4.5. Explainability

The detector loads with attn_implementation="eager" to materialize attention tensors. From the last layer's multi-head attention, we compute a CLS-to-patch attention rollout summary: mean/max CLS attention and attention entropy over patches. If tensors are unavailable (SDPA path, non-finite values, or empty), explainability status is unavailable with empty metrics — no fabricated metrics are substituted. This is **not** Grad-CAM; it is a CLS-token attention rollout derived from genuine ViT attention weights.

4.6. Authentication, Audit, and Rate Limiting

Authentication (Milestone 12): Argon2id (via argon2-cffi), JWT HS256 with exp/iat/iss/sub/role, 30-min expiry, HttpOnly/SameSite=Lax cookie + Bearer header fallback. Auth fails closed (503) when signing secret absent or <32 chars.

Audit (Milestone 14): Dedicated short-lived session per event; immediate commit; failures logged and swallowed (availability over atomicity). Bounded event types; dynamic strings clamped and charset-normalized. Never stored: passwords, tokens, cookies, image bytes, raw request bodies, EXIF metadata.

Rate limiting (Milestone 12): In-memory sliding window, per-process (single uvicorn worker). Keys: bounded prefixes + normalized UUID/IP only. Login 5/min per socket peer; scans 10/min; default 120/min. Distributed deployment would require Redis or gateway-level limiter.

5. Implementation

5.1. Detector Adapter

DetectorAdapter is a thread-safe singleton loading ViTForImageClassification and ViTImageProcessor once per process. Device selection: auto tries CUDA then CPU; cpu or cuda forces device. Inference runs with torch.inference_mode(), output_attentions=True. Logit → sigmoid → fake_probability $\in [0, 1]$. Latency measured with torch.cuda.synchronize() on GPU. Health endpoint reports detector_loaded, cuda_available, gpu_name, load_error.

Configuration (pinned):

```
model_id = "buildborderless/CommunityForensics-Dee
model_revision = "ac6ee457bea904a3730657541074517
expected_num_labels = 1
expected_shortest_edge = 440
expected_crop = 384
attn_implementation = "eager"
```

Config mismatch raises ConfigMismatchError at load time.

5.2. Scan Pipeline

1) POST /api/scans (authenticated, CSRF, rate-limited).
2) Validate MIME via decode (not header); decode via PIL; compute SHA-256.

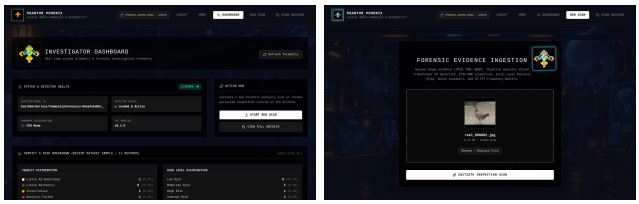


Figure 2: Investigator dashboard (left) and upload view with real-image preview (right).

3) Run detector inference → `ModelOutput`.
4) Run all forensic analyzers → `Indicator` list.
5) `aggregate_assessment` → normalized evidence + verdict + risk + summary.
6) Persist `Scan` row with full `full_result` JSON.
7) Return `ScanCreateResponse` with `scan_id`, `status`, `result`.

Database errors during persistence are logged but do not fabricate success; the computed result is still returned to the client.

5.3. Frontend Architecture

Single-file `App.tsx` (1,431 lines) with hash-based routing (`#landing`, `#login`, `#dashboard`, `#upload`, `#history`, `#scan-{id}`). Views: Landing, Login, Dashboard, Upload (drag-drop, preview), Results (verdict band, evidence cards, metadata table, artifact gallery, technical-details collapsible), Archive (scan list), PDF download. Internal `PhoenixMascot` SVG component with states: `idle`, `investigating`, `reassuring`. Design tokens in `index.css`: monospace typography, `.bg-*` background classes, amber primary CTA, CSS keyframe animations only.

5.4. PDF Report Generation

`report.py` uses ReportLab `platypus` to generate a multi-page PDF dossier containing: scan metadata, input details, verdict/risk/rationale, model output, forensic indicator results, evidence items, explainability metrics, and embedded artifact images (ELA, noise, FFT). Served via `GET /api/scans/{scan_id}/report.pdf` (authenticated).

6. Evaluation Methodology and Results

6.1. Dataset Construction

The PHANTOM PHOENIX evaluation manifest (`datasets/manifests/dataset_manifest.csv`) contains 6,000 images (3,000 real, 3,000 AI-generated) sourced from OpenFake `core` configuration. Stratified random sampling by `(label, generator_family)` with fixed seed 42 yielded:

TABLE 2: Locked test-set evaluation (922 samples, threshold 0.05). CI: 95% normal approximation for accuracy.

Metric	Value
Accuracy	0.7538 [0.7260, 0.7816]
Precision (AI)	0.9765
Recall (AI)	0.5298
F1 Score (AI)	0.6869
ROC-AUC	0.8959
PR-AUC	0.9163 (baseline, threshold-independent)
False Positive Rate	0.0133 (6 / 452)
False Negative Rate	0.4702 (221 / 470)
Balanced Accuracy	0.7583
Confusion Matrix	[[446, 6], [221, 249]]

TABLE 3: Threshold comparison on locked test set (922 samples).

Threshold	Accuracy	Precision	Recall	F1	FPR	FNR
0.50 (baseline)	0.6562	0.9935	0.3277	0.4928	0.0022	0.6723
0.20 (candidate)	0.6941	0.9845	0.4064	0.5753	0.0066	0.5936
0.05 (selected)	**0.7538**	**0.9765**	**0.5298**	**0.6869**	0.0133	0.4702

Split	Real	AI-Generated	Total
Train	2,099	2,089	4,188
Validation	449	441	890
Test	452	470	922

Zero SHA-256 duplicates within or across splits; zero missing/corrupt files (verified by `scripts/audit_physical_dataset.py`).

Test-set generator families (AI): `z-image-turbo` (128), `flux.2-klein-9b` (85), `illustrious` (61), `gpt-image-1.5` (59), `midjourney-7` (34), `veo-3` (21), `wan-video-2.5` (13), and 13 minor families ($n \leq 10$). Real test sources: ImageNet (304), DOCCI (148) distinct from training real sources (LAION, Pexels).

6.2. Evaluation Protocol

Model & Preprocessing.. `buildborderless/CommunityForensics-DeepfakeDet-ViT` @ revision `ac6ee457bea904a37306575410745179 3b56db00`, `attn_implementation="eager"`. ViTImageProcessor: shortest edge 440, center crop 384, CLIP normalization.

Threshold Selection (Validation).. Swept thresholds 0.050.95 on validation split (890 samples). Selected threshold maximizing balanced accuracy: **0.05**. Validation at 0.05: accuracy 0.7258, precision 0.9758, recall 0.4580, F1 0.6235, FPR 0.0111.

Frozen Test Evaluation.. The selected threshold (0.05) was applied once to the locked test split (922 samples) without further tuning.

6.3. Results

The selected threshold (0.05) increases AI recall from 32.77% to 52.98% while maintaining FPR at 1.33% (6 false positives on 452 real images). Production thresholds (0.80 /

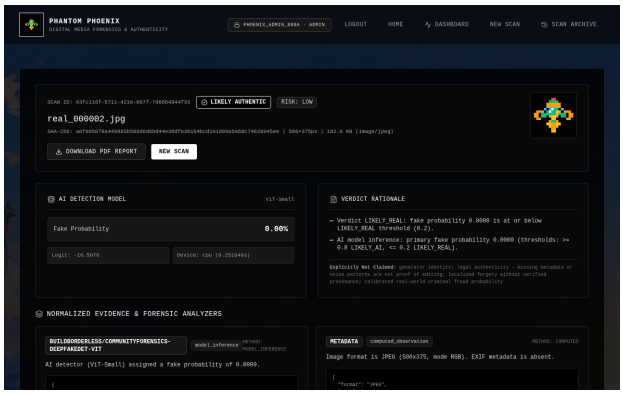


Figure 3: Results overview: verdict band (LIKELY_REAL), forensic score, structured evidence cards replacing raw JSON.

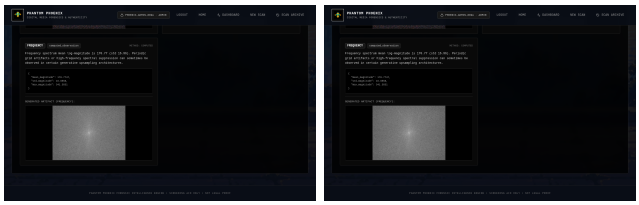


Figure 4: Metadata & evidence detail (left) and forensic artifact gallery (ELA, noise, FFT) (right). Artifacts are served via authenticated `/api/artifacts/` endpoints.

0.20) are distinct from evaluation thresholds and prioritize screening conservativeness.

Inference Latency.. On CPU (deployment environment): mean 39.7 ms, median 27.5 ms, total 36.6 s for 922 images (batch size 1).

System Verification Tests.. Frontend: TypeScript strict build (0 errors), Vite production build (279 kB JS, 82 kB gzip), oxlint (5 pre-existing warnings), Playwright E2E smoke (4/4 passed: landing gate, hash routing, protected redirect, generic login failure). Backend: 156 unit tests pass (166 with DB). PostgreSQL integration tests not rerun against the valued `pp13` database (fixture deletes scan rows). Docker Compose config valid; `git diff -check` clean.

7. Case Walkthrough

1) **Landing.** User arrives at `https://host/#landing` (Fig. 1); clicks `Investigator Dashboard` → routed to sign-in.
2) **Authentication.** Credentials submitted to `POST /api/auth/login`; HttpOnly JWT cookie set; redirected to `#dashboard` (Fig. 2 left).
3) **Upload.** User drags `real_000002.jpg` (900×700 PNG, 93 KB); preview renders; submits (Fig. 2 right).
4) **Inference.** Backend validates, decodes, runs ViT-Small detector (40 ms), four forensic analyzers, aggregates evidence, determines verdict, persists scan.
5) **Results.** Browser navigates to `#scan-{id}`; renders verdict, score, evidence cards, metadata table, artifact gallery (Fig. 3, 4). Raw JSON available in collapsible `Technical Details`.
6) **PDF.** User clicks `Download PDF Report` → `GET /api/scans/{id}/report.pdf` returns 400 KB `application/pdf` (%PDF-1.4).
7) **Archive.** User visits `#history` → paginated scan list from `GET /api/scans`.
8) **Logout.** Session cookie cleared; subsequent `/api/auth/me` returns 401; user routed to `#login`.

All steps verified against the running `pp13` staging stack.

8. Limitations, Threats to Validity, and Ethical Considerations

Dataset and Evaluation Limitations..

- The 6,000-image manifest derives from OpenFake `core` v1; generator-family labels reflect provider metadata and may contain misattributions.
- Minor generator families ($n < 5$) appear in both train and test splits; true unseen-generator generalization requires holding out entire major families (not done).
- Resolution/format confounding exists: Decision Tree on 6 non-semantic features achieves 92% CV accuracy on 200-image subset, indicating measurable dataset-source shortcut features [13].
- The validation threshold selection (maximizing balanced accuracy at 0.05) was performed on the same validation split used for hyperparameter awareness; the test evaluation is frozen but the selection criterion itself is not independently validated on a third hold-out set.
- Historical pilot metrics (50-image OpenFake pilot, accuracy 0.62, recall 0.24 at 0.50) are reported in `PHASE2_PILOT_ERROR_ANALYSIS.md` but are not representative of the full 922-sample test evaluation.
- No ablation study isolates the contribution of forensic analyzers to verdict accuracy; evidence fusion improvement claims are not substantiated by controlled baselines.

System Limitations..

- Single-worker in-memory rate limiting; horizontal scaling requires Redis/gateway.
- JWT tokens are snapshots; account disable/role revocation takes effect at token expiry (≤30 min); no server-side session store or denylist.
- Audit trail is not a tamper-proof ledger (DB admin can modify); availability prioritized over forced atomicity.

- Explainability is attention rollout, not gradient-based (Grad-CAM); may not localize generator-specific artifacts.
- Production thresholds (0.80/0.20) are provisional screening defaults, not calibrated probabilities.
- Image-only; no video, audio, or multimodal analysis.

Ethical Considerations..

- The platform is a *screening aid*, not a forensic proof system. Mandatory `not_claimed` disclaimers are part of every verdict.
- False positives (real images flagged as AI) carry reputational and evidentiary risk; production threshold 0.80 minimizes this at the cost of recall.
- No biometric data processed; EXIF/XMP stripped from stored artifacts; no PII in audit logs.
- Model trained on Community Forensics dataset (CC-BY-NC-4.0 subsets); deployment respects license terms.

9. Conclusion and Future Work

PHANTOM PHOENIX delivers a complete, auditable forensic screening pipeline: ViT-Small detector (Community Forensics), classical forensic analyzers, normalized evidence schema, deterministic verdict engine, authenticated persistence, audit logging, and PDF dossiers. On the locked 922-sample test split, the frozen validation-selected threshold achieves 75.38% accuracy, 52.98% AI recall, 1.33% FPR, and 0.8959 ROC-AUC.

Future work:

- Calibrated probability outputs (temperature scaling, isotonic regression) to replace provisional thresholds.
- Unseen-generator hold-out evaluation (entire major families withheld from training).
- Ablation studies quantifying forensic analyzer contribution to verdict accuracy.
- Gradient-based explainability (Integrated Gradients, attention-flow) with artifact generation.
- Multi-modal extension: video frame analysis, audio-visual consistency.
- Distributed rate limiting and audit hardening for multi-replica deployments.
- Formal red-team evaluation of adversarial robustness.

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